

최적궤도제어

기말고사 20011 년 가을

1. $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} + \mathbf{f}(t), \quad \mathbf{x}(0) = \mathbf{x}_0$

$\mathbf{f}(t)$ is a known forcing function.

$$J = \frac{1}{2}(\mathbf{x}^T \mathbf{Q} \mathbf{x})_{t=t_f} + \frac{1}{2} \int_0^{t_f} (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt$$

t_f is known. Show that

$$\mathbf{u}(t) = -\mathbf{R}^{-1} \mathbf{B}^T (\mathbf{k} \mathbf{x} + \mathbf{r})$$

$$\dot{\mathbf{r}}(t) = (\mathbf{k} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T - \mathbf{A}^T) \mathbf{r} - \mathbf{k} \mathbf{f}(t)$$

$\mathbf{k}$ satisfies the Riccati equation.

$$H = g + \lambda^T f$$

$$\frac{\partial H}{\partial \mathbf{u}} = \mathbf{R} \mathbf{u} + \mathbf{B}^T \lambda = 0$$

$$\mathbf{u} = -\mathbf{R}^{-1} \mathbf{B}^T \lambda$$

$$\dot{\lambda} = -\left(\frac{\partial H}{\partial \mathbf{x}}\right)^T = -\mathbf{Q} \mathbf{x} - \lambda^T \mathbf{A}$$

$$\lambda = \mathbf{k} \mathbf{x} + \mathbf{r}$$

2.

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -a x_2 - b x_1 + u$$

$$J = \frac{1}{2} \int_0^{t_f} [t(c_1 x_1^2 + c_2 x_2^2)] dt$$

where a, b, c_1 and c_2 are constant. Find the optimal control law on singular intervals and on regular intervals.

3. The system equation are

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = u_1$$

$$\dot{x}_3 = x_4, \quad \dot{x}_4 = u_2$$

$\mathbf{x}(0)$ are given and $\mathbf{x}(t_f)$ are free.

There is a state equality constraint as:

$$S(\mathbf{x}) = x_1^2 + x_3^2 - 1 = 0$$

The performance index to be minimizes is

$$J = \int_0^{t_f} (u_1^2 + u_2^2) dt, \quad t_f \text{ is fixed}$$

Explain how to solve this problem.

4.

Explain the algorithm of a direct method and explain also the algorithm of an indirect method, in order to numerically solve optimal control problems.

$$1. \dot{x} = Ax + Bu + f \quad x(0) = x_0$$

$$J = \frac{1}{2}(x^T Q x)_{t=t_f} + \frac{1}{2} \int_0^{t_f} (x^T Q x + u^T R u) dt$$

$$H = g + \lambda^T f = \frac{1}{2} x^T Q x + \frac{1}{2} u^T R u + \lambda^T A x + \lambda^T B u + \lambda^T f$$

$$\frac{\partial H}{\partial u} = R u + \lambda^T B$$

$$u = -R^{-1} B^T \lambda = -R^{-1} B^T (kx + r)$$

$$\dot{\lambda} = -\left(\frac{\partial H}{\partial x}\right)^T = -(x^T Q + \lambda^T A)^T = -Qx - A^T \lambda$$

$$\dot{\lambda} = -Qx - A^T \lambda$$

$$\lambda = kx + r$$

$$\dot{\lambda} = \dot{k}x + k\dot{x} + \dot{r}$$

$$\dot{r} = \dot{\lambda} - \dot{k}x - k\dot{x}$$

$$= -Qx - A^T(kx + r) - \dot{k}x - kAx - kBu - kf$$

$$= -kf - Qx - A^T kx - A^T r - \dot{k}x - kAx + kBR^T B^T (kx + r)$$

$$= -kf + (kBR^T B^T - A^T) r - Qx - A^T kx - \dot{k}x - kAx + kBR^T B^T kx$$

$$= (kBR^T B^T - A^T) r - kf + \underbrace{(-Q - A^T k - \dot{k} - kA + kBR^T B^T k)}_{\text{Riccati eqn'}} x$$

$$\dot{k} + kA - kBR^T B^T k + A^T k + Q = 0$$

Riccati eqn'

$$2. \dot{x}_1 = x_2$$

$$\dot{x}_2 = -ax_1 - bx_1 + u$$

$$J = \frac{1}{2} \int_0^t (t c_1 x_1^2 + t c_2 x_2^2) dt$$

$$H = g + \Lambda^T f = \frac{1}{2} t c_1 x_1^2 + \frac{1}{2} t c_2 x_2^2 + \Lambda_1 x_2 - \Lambda_2 a x_1 - \Lambda_2 b x_1 + \Lambda_2 u$$

$$\frac{\partial H}{\partial u} = \Lambda_2 = 0$$

$$\Lambda_2 = 0$$

$$\dot{\Lambda}_1 = -\frac{\partial H}{\partial x_1} = -\left(t c_1 x_1 + \cancel{t c_2} - \cancel{\Lambda_2 a} - \cancel{\Lambda_2 b} \right)$$

$$= -t c_1 x_1$$

$$\dot{\Lambda}_2 = -\frac{\partial H}{\partial x_2} = -t c_2 x_2 - \Lambda_1$$

$$\dot{\Lambda}_2 = -t c_2 x_2 - \Lambda_1$$

$$= -c_2 x_2 - t c_2 \dot{x}_2 - \dot{\Lambda}_1$$

$$= -c_2 x_2 - t c_2 (-ax_1 - bx_1 + u) + t c_1 x_1 = 0$$

$$\frac{t c_1 x_1 - c_2 x_2}{t c_2} = -ax_1 - bx_1 + u$$

$$u = ax_1 + bx_1 + \frac{t c_1 x_1 - c_2 x_2}{t c_2}$$

$$t=0 \text{ or } c_2=0 \text{ singular}$$

2014 풀이 참조

2.

$$m = \text{mass}$$

$$-M \leq \dot{m} \leq 0 \Rightarrow -M \leq u \leq 0$$

$$x(t_f) = 0$$

g

$$T = -k\dot{m}$$

$$m\ddot{x} = -gm + T$$

$$m\ddot{x} = -gm - k\dot{m}$$

$$\ddot{x} = -g - \frac{k}{m}u$$

$$x_1 = x$$

$$\dot{x}_1 = \dot{x}_2$$

$$x_2 = \dot{x}$$

$$\dot{x}_2 = -g - \frac{k}{m}u$$

$$J = \int_{t_0}^{t_f} 1 dt$$

$$H = g + \lambda^T f = 1 + \lambda_1 x_2 - \lambda_2 g - \lambda_2 \frac{k}{m}u$$

$$\frac{\partial H}{\partial u} = -\lambda_2 \frac{k}{m}$$

← control이 없으면 singular interval

$$\dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = 0$$

$$\lambda_1 = \text{constant}$$

$$\dot{\lambda}_2 = -\left(\frac{\partial H}{\partial x_2}\right) = -\lambda_1$$

$$\lambda_2 = -\lambda_1 t + \lambda_{2,0}$$

2-1) solve the problem using Pontryagin minimum principle

Pontryagin minimum principle은

H이 최소가 되는 u(t)를 찾아야 함으로

$$H = 1 + \lambda_1 x_2 - \lambda_2 g - \lambda_2 \frac{k}{m}u$$

위식에서 λ_2 와 u의 부호가 같으면 H이 최소가 되고 u의 부호가 $-M \leq \dot{m} \leq 0$ 이므로

$$u \begin{cases} 0 & \text{---} & \lambda_2 > 0 \\ -M & \text{---} & \lambda_2 < 0 \\ \text{Undetermined} & \lambda_2 = 0 \end{cases}$$

즉 $\lambda_2 = -\lambda_1 t + \lambda_{2,0}$ 이다

1. $\dot{x}_1(t) = x_2(t)$ $x(t) = \text{given}$
 $\dot{x}_2(t) = -x_1(t) + u(t)$ $t_f = \text{specified}$
 $x(t_f) = \text{free}$

$$J = \int_0^{t_f} \left(\frac{1}{2} x_1^2 + \frac{1}{2} u^2 \right) dt$$

constraint $\Rightarrow -1 \leq u(t) \leq 1$ $u \in \mathbb{R}$

$$H = g + \lambda^T f = \frac{1}{2} x_1^2 + \frac{1}{2} u^2 + \lambda_1 x_2 - \lambda_2 x_2 + \lambda_2 u$$

if no constraint

$$\frac{\partial H}{\partial u} = u + \lambda_2 = 0$$

$$u = -\lambda_2$$

$$\Rightarrow -1 \leq \lambda_2 \leq 1$$

Pontryagin's minimum principle에 의하여 H를 최소화하는 u를 찾는 것

H가 최소가 되려면 $\lambda_2 u$ 가 최소가 되어야 함으로

u와 λ_2 의 sign이 반대가 되어야 하고 크기는 최대값인 1 or -1을 가진다.

u가 1이 클수록 H가 작아진다. λ_2 가 0이면 그 크기는 정해지지 않으므로 이때에 같이 optimal control을 찾는다

$$u = \begin{cases} 1 & \lambda_2 < -1 \\ -\lambda_2 & -1 \leq \lambda_2 \leq 1 \\ -1 & \lambda_2 > 1 \end{cases}$$

Undetermined when $\lambda_2 = 0$

2. $m = \text{mass}$

$$-1 \leq \ddot{m} \leq 0$$

$$g$$

$$T = -km$$

$$m\ddot{x} = -gm + T$$

State eq $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -g + \frac{-km}{m} \end{cases}$

$$u = m\ddot{x}_2 + gm$$

2-1) $J = \int_0^{t_f} 1 dt$

$$H = g + \lambda^T f = 1 + \lambda_1 \dot{x}_2 - \lambda_2 g - \lambda_2 \frac{u}{m}$$

Pontryagin's minimum principle은 H가 최소가 되는 u를 찾는 것이므로

λ_2 의 부호와 u의 부호가 반대가 되어야 함, 크기는 최대

$$u(t) = \begin{cases} T & \lambda_2 > 0 \\ -T & \lambda_2 < 0 \\ \text{undetermined} & \lambda_2 = 0 \end{cases}$$

$$J = x^2(1) + \int_0^1 \frac{1}{2} u^2(t) dt$$

앞면에서 계속

4. $\dot{x}(t) = -x(t) + u(t)$
 $x(0) = 4$
 $t_0, t_f = \text{given}$

$$H = g + \lambda^T f$$

$$= \frac{1}{2} u^2 - \lambda_1 x + \lambda_1 u$$

① guess $u(t)$ $\Leftarrow$ t_f fixed 이므로 $u(t)$ 를 guess

$\Downarrow$

② $\dot{x} = -x(t) + u(t)$ $\Leftarrow$ 이로부터 $x(t_0)$ 부터 $x(t)$ 를 만든다

③ $\dot{\lambda}(t) = \left(\frac{\partial H}{\partial x} + \lambda^T \frac{\partial f}{\partial x} \right)_{t=t_f} \Leftarrow$ 이로부터 $\lambda(t_f)$ 를 구한다

$$\dot{\lambda}(t) = \frac{\partial H}{\partial x} = 2x x(1) \Leftarrow \text{②번에서 구한 } x(1) \text{ 을 } x(t_f) \text{ 로 치환한다.}$$

④ $\dot{\lambda} = -\frac{\partial H}{\partial x}$ 을 이용하여 $\lambda(t_f)$ 부터 Backward Integration을 한다

$$\dot{\lambda}_1 = -(-\lambda_1) \Rightarrow \dot{\lambda}_1 = \lambda_1, \lambda(t_f) = 2x x(t_f)$$

$\rightarrow \lambda(t)$ 의 history를 구한다

⑤ $\frac{\partial H}{\partial u} = 0$ optimality condition을 계산하여 그 크기가 만족할만한 수준인지 구한다
 $\Delta H^k = \frac{\partial H}{\partial u} < \epsilon \Leftarrow$ ϵ 이 너무 크면 oscillation을 생기기 쉽고
 ϵ 이 너무 작으면 시보이팅이 일어나 정확한 값을 찾아야 한다

⑥ 만약 조건을 만족하지 못하면 $u(t)$ 를 update 한다

$$u^{k+1}(t) = u(t) - \alpha \frac{\partial H}{\partial u} \Leftarrow \alpha > 0$$

여기 $-\alpha \frac{\partial H}{\partial u} = \Delta u(t)$ 이며 $\Delta u(t)$ 를 δJ 가 정해지
 방향으로 설정하기 위해이다

⑦ 새로 얻은 $u^{k+1}(t)$ 를 가지고 다시 ②번부터 반복한다

만약 t_f 가 free이면 t_f 도 guess 해야 하며 초기에 $u(t)$ 나 t_f 를 guess 하고
 $\Delta u = -K[\] - \eta[\]$ 의 형태로 구하고 η 를 Δu 를 판정할 때 쓴다

계속

optimality condition

$$2-2 \quad \frac{\partial H}{\partial u} = -\lambda_2 \frac{k}{m} m < 0$$

control의 값을 정의하기 못함

$\Rightarrow$ Singular interval

$$-\lambda_2 \frac{k}{m} = 0$$

$$\lambda_2 = 0$$

이 결과를 알기 Pontryagin Minimum principle에 한 결과를 대입하면
전 t 의 구간에서 $-u$ 를 정의할 수 없다.

근바라기

Singular interval 구간 찾기 위해 계속 계산을 하면

$$\begin{cases} \lambda_2 = 0 \\ \dot{\lambda}_2 = -\lambda_1 \leftarrow \text{from costate eqn'} \\ \dot{\lambda}_2 = 0 \end{cases}$$

$$\lambda_2 = -\lambda_1 t + \lambda_{20}$$

$$\begin{cases} -\lambda_1 = 0 \\ \lambda_1 = \text{constant} \end{cases}$$

$$\lambda_1 = \frac{\lambda_{2,0} - \lambda_2}{t}$$

$$\ddot{\lambda}_2 = 0$$

$\downarrow$

계속해볼때도 $-u$ 의 형태를 찾을 수 없다.

$\Rightarrow$ Pontryagin minimum principle로 돌아간다.

과목명	
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성 적	
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$$\delta J_a = \frac{\partial \Phi}{\partial t_1} \delta t_1^+ + \frac{\partial \Phi}{\partial t_1} \delta t_1^- + \Phi \delta t_1 - \Lambda^T (\delta x_{t_1} - \delta \dot{x}_{t_1} \delta t_1) + \Lambda^T (\delta x_{t_1^+} - \delta \dot{x}_{t_1^+} \delta t_1^+)$$

$$+ \int_0^{t_1^-} \left[\left(\frac{\partial H}{\partial x} + \Lambda^T \right) \delta x + \left(\frac{\partial H}{\partial u} \delta u \right) + \left(\frac{\partial H}{\partial \lambda} - \dot{x} \right) \delta \lambda + \delta \Lambda^T f \right] dt + [H + \Lambda^T (f - \dot{x})] \delta t_1$$

$$+ \int_{t_1^+}^0 \left[\left(\frac{\partial H}{\partial x} + \Lambda^T \right) \delta x + \left(\frac{\partial H}{\partial u} \delta u \right) + \left(\frac{\partial H}{\partial \lambda} - \dot{x} \right) \delta \lambda + \delta \Lambda^T f \right] dt - [H + \Lambda^T (f - \dot{x})] \delta t_1$$

$$\delta J_a = \left[\frac{\partial \Phi}{\partial t_1^+} - \Lambda^T \delta \dot{x}_{t_1^+} \right] \delta t_1^+ + \Phi \delta t_1$$

$$\left[\frac{\partial \Phi}{\partial t_1^-} + \Lambda^T \delta \dot{x}_{t_1^-} \right] \delta t_1^- + \Lambda^T (-\delta x_{t_1^-} + \delta x_{t_1^+})$$

$$+ \int_0^{t_1^-} \left[\left(\frac{\partial H}{\partial x} + \Lambda^T \right) \delta x + \left(\frac{\partial H}{\partial u} \delta u \right) + \left(\frac{\partial H}{\partial \lambda} + \dot{x} \right) \delta \lambda + \delta \Lambda^T f \right] dt + [H + \Lambda^T (f - \dot{x})] \delta t_1$$

$$+ \int_{t_1^+}^0 \left[\left(\frac{\partial H}{\partial x} + \Lambda^T \right) \delta x + \left(\frac{\partial H}{\partial u} \delta u \right) + \left(\frac{\partial H}{\partial \lambda} + \dot{x} \right) \delta \lambda + \delta \Lambda^T f \right] dt - [H + \Lambda^T (f - \dot{x})] \delta t_1$$

$$\Lambda^T = - \frac{\partial H}{\partial x}$$

$$\dot{x} = + \frac{\partial H}{\partial \lambda}$$

$$\frac{\partial H}{\partial u} = 0$$

$$\frac{\partial \Phi}{\partial t} + H(t^+) = \Phi(t_-) \Leftarrow \Phi \text{가 } t \text{에 대해 explicit하지 않음}$$

$H = \text{continuous}$

앞면에서 계속

$$3. \quad \dot{x} = u \quad x(0) = x_0, \quad x(1) = 0 \\ x(t_1) = 0$$

$$J = \int_0^1 g(x, u) dt$$

$$(x(t_1^+) - x(t_1^-) - 1) = 0$$

$$g + \lambda^T f$$

$$J_a = \underbrace{V^T \{x(t_1^+) - x(t_1^-) - 1\}}_{\frac{1}{\Phi}} + \int_0^1 (H + \lambda^T (f - \dot{x})) dt$$

$$f'g = [\quad] -$$

$$(fg)' = f'g + fg'$$

$$\delta J_a = \frac{\partial \Phi}{\partial t_1^+} \delta t_1^+ + \frac{\partial \Phi}{\partial t_1^-} \delta t_1^- + \Phi \delta t_1$$

$$+ \int_0^{t_1^-} \left[\frac{\partial H}{\partial x} \delta x + \frac{\partial H}{\partial u} \delta u + \frac{\partial H}{\partial \lambda} \delta \lambda + \delta \lambda^T f - \delta \lambda^T \dot{x} - \lambda^T \delta \dot{x} \right] dt + [H + \lambda^T (f - \dot{x})] \delta t_1$$

$$+ \int_{t_1^+}^1 \left[\frac{\partial H}{\partial x} \delta x + \frac{\partial H}{\partial u} \delta u + \frac{\partial H}{\partial \lambda} \delta \lambda + \delta \lambda^T f - \delta \lambda^T \dot{x} - \lambda^T \delta \dot{x} \right] dt - [H + \lambda^T (f - \dot{x})] \delta t_1$$

$$= \frac{\partial \Phi}{\partial t_1^+} \delta t_1^+ + \frac{\partial \Phi}{\partial t_1^-} \delta t_1^- + \Phi \delta t_1 +$$

$$+ \left[-\lambda^T \delta x \Big|_0^{t_1^-} \right] + \int_0^{t_1^-} \left[\left(\frac{\partial H}{\partial x} + \lambda^T \right) \delta x + \left(\frac{\partial H}{\partial u} \right) \delta u + \left(\frac{\partial H}{\partial \lambda} - \dot{x} \right) \delta \lambda + \delta \lambda^T f \right] dt + [H + \lambda^T (f - \dot{x})] \delta t_1$$

$$+ \left[-\lambda^T \delta x \Big|_{t_1^+}^1 \right] + \int_{t_1^+}^1 \left[\left(\frac{\partial H}{\partial x} + \lambda^T \right) \delta x + \left(\frac{\partial H}{\partial u} \right) \delta u + \left(\frac{\partial H}{\partial \lambda} - \dot{x} \right) \delta \lambda + \delta \lambda^T f \right] dt - [H + \lambda^T (f - \dot{x})] \delta t_1$$

$$= \frac{\partial \Phi}{\partial t_1^+} \delta t_1^+ + \frac{\partial \Phi}{\partial t_1^-} \delta t_1^- + \Phi \delta t_1 - \lambda^T \delta x(t_1^-) + \lambda^T \delta x(t_1^+)$$

$$+ \int_0^{t_1^-} \left[\left(\frac{\partial H}{\partial x} + \lambda^T \right) \delta x + \left(\frac{\partial H}{\partial u} \right) \delta u + \left(\frac{\partial H}{\partial \lambda} - \dot{x} \right) \delta \lambda + \delta \lambda^T f \right] dt + [H + \lambda^T (f - \dot{x})] \delta t_1$$

$$+ \int_{t_1^+}^1 \left[\left(\frac{\partial H}{\partial x} + \lambda^T \right) \delta x + \left(\frac{\partial H}{\partial u} \right) \delta u + \left(\frac{\partial H}{\partial \lambda} - \dot{x} \right) \delta \lambda + \delta \lambda^T f \right] dt - [H + \lambda^T (f - \dot{x})] \delta t_1$$

or
 $\delta x_{t_1^\pm} = \delta x(t_{1^\pm}) + \dot{\delta x}_{t_1^\pm} \cdot \delta t_{1^\pm} \Rightarrow \delta x(t_{1^\pm}) = \delta x_{t_1^\pm} - \dot{\delta x}_{t_1^\pm} \cdot \delta t_{1^\pm}$